

M ASWIN

(+91) 6379482324 | aswinmmsa@gmail.com | LinkedIn | GitHub

EDUCATION

Vellore Institute of Technology – AP
B.Tech in Computer Science & Engineering
Sri Chaitanya Techno School
Class XII (CBSE)

Sep 2023 – Present
GPA: 8.39/10
Jun 2021 – Apr 2022

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java, SQL
- **Frameworks & Libraries:** FastAPI, LangChain, LangGraph, MCP, Scikit-learn, Pandas, OpenCV, TensorFlow, PyTorch, Neattext
- **Tools & Platforms:** Git, MySQL, MATLAB (Simulink), ThingSpeak IoT, MIT App Inventor
- **Core Competencies:** Artificial Intelligence (AI), Data Science, Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Data Analysis, Statistics, Feature Engineering, Healthcare AI, Scientific Literature Analysis

PROJECTS

EmotionsBasedOnTexts (2025) [Link](#)
Python, Scikit-learn, Pandas, Neattext, NLP

- Developed a **multilingual emotion detection system** using **Python** to classify emotions such as **joy, sadness, anger, and fear** from user text.
- Implemented automatic **language detection and translation** to support multilingual text analysis.
- Performed **data preprocessing** and **feature engineering** using NLP techniques including **tokenization, stopwords removal, and lemmatization**.
- Extracted features using **TF-IDF** and trained **Machine Learning** models (**Logistic Regression, SVM, Random Forest**) for multilingual emotion classification.
- Evaluated model performance using **accuracy, precision, recall, and F1-score**, demonstrating strong results for a **Data Science** application on CPU-based deployment.

AgenticHealthAI – AI-Powered Healthcare Assistant (2026) [Link](#)
Python, FastAPI, LangChain, MCP, GROQ, MySQL

- Developed a **Python-based AI healthcare assistant** providing symptom triage, clinical insights, and nearby doctor discovery.
- Built a **FastAPI** backend with REST APIs for authentication, chat interaction, medical profile management, and conversation history using **MySQL**.
- Integrated a **GROQ-powered medical AI agent** with an MCP server to retrieve **biomedical research papers and scientific literature**, enabling evidence-based clinical summaries and clinical decision support.
- Implemented geolocation-based doctor discovery using the **Haversine formula** and developed a responsive interface with TailwindCSS.

Multilingual OCR, Translation & TTS System (2025) [Link](#)
Python, PaddleOCR, OpenCV, FastAPI, Groq LLaMA 4 Scout, Google Translator API, PlayAI TTS

- Developed an end-to-end **Python-based OCR, Translation, and Text-to-Speech** pipeline for extracting multilingual text from images and converting it into speech.
- Leveraged **Groq LLaMA 4 Scout (Vision LLM)** to automatically detect the language of extracted text using the ISO 639-1 standard.
- Implemented **PaddleOCR** and **OpenCV** for accurate multilingual text recognition from uploaded images and real-time camera captures.
- Integrated the **Google Translator API** for multilingual translation and **Groq PlayAI TTS** for high-quality speech synthesis.
- Built a real-time image processing pipeline using **ESP32-CAM** and OpenCV for low-latency text extraction and processing.

EXPERIENCE

Code Crux Club – Competitive Programming Team (2025–Present)

- * Solved 200+ problems across LeetCode.
- * Active in mock contests & ICPC-style challenges.